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Design and Control of a Subsurface Untethered Burrowing Robot (SUBOT) for Deploying Exploration Sensors in Loose Sand Environments

Raed Alahmdi, Obadah Wali, and Mohammad Sait, Saudi Aramco PE&D - EXPEC ARC - KURC; Pavel Golikov, Saudi Aramco PE&D - EXPEC ARC - GPD

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Abstract

This paper presents the development and evaluation of a fully autonomous, untethered robotic system—referred to as SUBOT—designed for deploying seismic exploration sensors beneath the surface of loose sandy environments. SUBOT introduces a four-legged screw-driven locomotion design, allowing it to drill vertically into the sand without external tethering or anchoring mechanisms. Each leg of the robot is independently actuated using high-torque rotary motors, providing the mechanical force necessary for vertical propulsion. The mechanical structure of the robot, including a square-frame motor support and a cross-shaped chassis formed by L-beams, was carefully engineered for structural integrity, torque distribution, and compactness. In addition to mechanical considerations, the robot integrates a dual PID-based feedback control system, utilizing an inertial measurement unit (IMU) to autonomously stabilize its orientation during burrowing. This control system mitigates tilt and ensures consistent vertical motion even in highly granular environments. Experimental testing in a controlled sandbox environment demonstrated the robot's ability to autonomously burrow to depths of up to 75 cm, self-level in response to pitch and roll disturbances, and ascending without human intervention. The results highlight the system's potential for improving subsurface seismic sensor deployment, offering an automated alternative to manual placement in land exploration and environmental monitoring applications.

Introduction

The need for efficient, accurate, and autonomous subsurface data collection has driven significant interest in bio-inspired and field-deployable robotic systems. Among these, burrowing robots stand out due to their capacity to operate beneath the surface, offering access to regions that are difficult, dangerous, or impossible for humans or traditional equipment to reach. Burrowing robots have demonstrated promising potential across diverse applications including geotechnical engineering, underground inspection, planetary exploration, and environmental monitoring (Nagaoka et al. 2008; Tadami et al. 2017). In each of these domains, the ability to autonomously penetrate, stabilize, and navigate within granular or stratified subsurfaces can greatly enhance the quality and resolution of collected data.

One particularly valuable use case lies in seismic sensing, where obtaining accurate seismic data depends on the degree to which a sensor is coupled to the ground. Shallow or poorly placed geophones can suffer from noise interference, poor data fidelity, and inconsistent coupling that undermine the reliability of seismic surveys. For high-quality subsurface imaging—especially in the oil and gas sector—seismic sensors must be buried at controlled depths to ensure both mechanical contact and environmental noise reduction (Bland et al. 2001). Despite its critical role, sensor burial is often performed manually, which is labor-intensive, time-consuming, and exposes workers to hazardous conditions, particularly in remote or hostile terrain.

The pursuit of robotic alternatives for sensor deployment has led to multiple innovations. Early prototypes, such as the lunar screw-driven subsurface explorer, demonstrated the feasibility of drilling into loosely compacted sand at modest depths, with robots reaching up to 12 cm depth using approximately 1 N·m of torque (Nagaoka et al. 2009). Lichtenheldt et al. (2017) introduced a screw-driven burrowing robot capable of reaching 20 cm with 0.66 N·m torque, though their system was constrained by a narrow diameter and limited by power and control constraints. Abe et al. (2010) tested a contra-rotating screw system with torque levels around 0.1 N·m, though their evaluations focused on very shallow depths (<20 cm). These studies collectively demonstrate the viability of screw-based locomotion in loose sand, but are limited either by tethering, lack of autonomous control, or constrained mechanical stability.

A gap remains in the design of untethered, seismic sensor carrier, closed-loop-controlled systems that can both burrow to useful depths (≥ 50 cm) and maintain consistent orientation despite torque-induced disturbances. Building on our previous work in optimizing screw-body designs for digging in shallow, loose sand environments (Alahmadi et al. 2023), the present study addresses a critical gap by developing SUBOT—a self-contained robotic system capable of autonomously burrowing seismic sensors beneath sandy surfaces and returning without external input. The robot incorporates quad-leg screw-driven propulsion system that mimics certain principles from quadrotor drones in how torques are balanced and distributed across actuated limbs.

Moreover, this system integrates an inertial feedback-based PID control loop that dynamically stabilizes the robot's orientation during descent and ascent. This paper presents the design, modeling, control implementation, and experimental validation of SUBOT, and explores its potential implications for field-deployable subsurface sensing systems.

Mechanical Design

The mechanical design of SUBOT is fundamental to its ability to autonomously burrow and resurface in loose sand. The system must withstand substantial drilling forces, remain structurally stable during asymmetric torque conditions, and provide an effective interface between the propulsion mechanism and granular media. This section presents the mechanical components and structural architecture of the robot in detail, including the motor support system, bushing stabilization, leg configuration, and chassis design.

Motor Structure and Support Assembly

At the heart of SUBOT's burrowing capability are four high-torque DC motors, each capable of delivering up to 10 N·m torque. These motors are mounted within a support structure composed of a square-shaped aluminum frame with a circular cutout through which the motor housing is inserted. The motor casing is secured using 4M screws, ensuring attachment while allowing modular access for maintenance or replacement.

To reinforce the motor housing, four aluminum L-beams are mounted at each corner of the square motor support (Fig. 1). These beams extend outward to form part of the robot's load-bearing chassis. The connection between the motor support and L-beams is used distributing drilling-induced stress away from the motor shaft and preventing rotational misalignment under load.

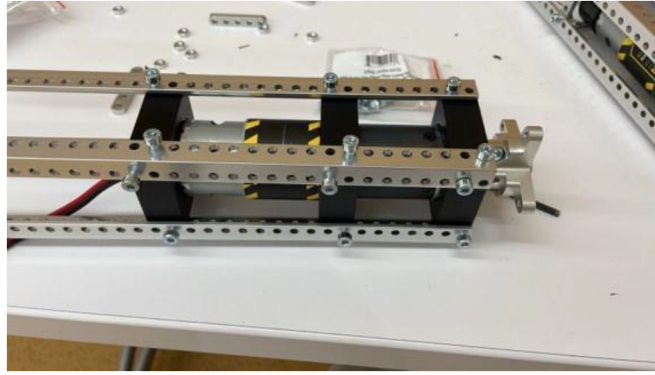


Figure 1—Motor support structure with square frame and L-beam attachments

Bushing Support and Radial Stability

To address vibration and lateral instability—particularly during high-speed rotation in non-uniform granular media—bushing supports were added between the motor shaft and leg body as shown in Fig.2. These bushings limit lateral deflections in the X–Y plane and reduce oscillatory forces that may degrade sensor readings or destabilize the robot. The bushings are made from polymers, ensuring wear resistance and minimal added weight.

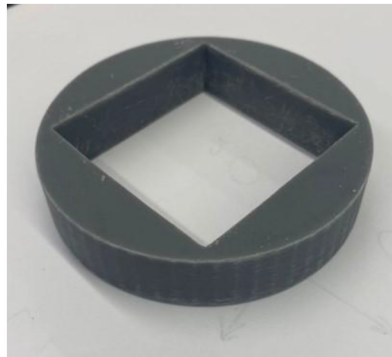


Figure 2—Bushing support showing its connection between the motor structure and the leg, reducing X and Y fluctuations.

Experimental observations revealed that the bushing assembly improved the stability of the leg's path during rotation. Without bushings, motors exhibited visible deviation under torque, leading to tilt accumulation and inefficient drilling. With bushings, such deviations were minimized, contributing to effective control loop behavior and accurate vertical penetration.

Leg Geometry and Rotational Configuration

SUBOT's propulsion relies on four screw-threaded legs, each approximately 30 cm in length and fabricated using PETG plastic via fused deposition modeling (FDM) 3D printing as shown in Fig. 3. Each leg integrates helical threads designed to engage with granular particles during rotation, creating thrust in the vertical direction.

To maintain torque equilibrium and prevent rotational drift, the legs are actuated in a diagonally symmetric fashion: two legs rotate clockwise (CW), and two counter-clockwise (CCW). Specifically, each pair of diagonal legs is set to the same rotational direction—this configuration mirrors the torque balancing strategy used in quadcopters and ensures the net yaw torque of the robot remains near zero during operation.

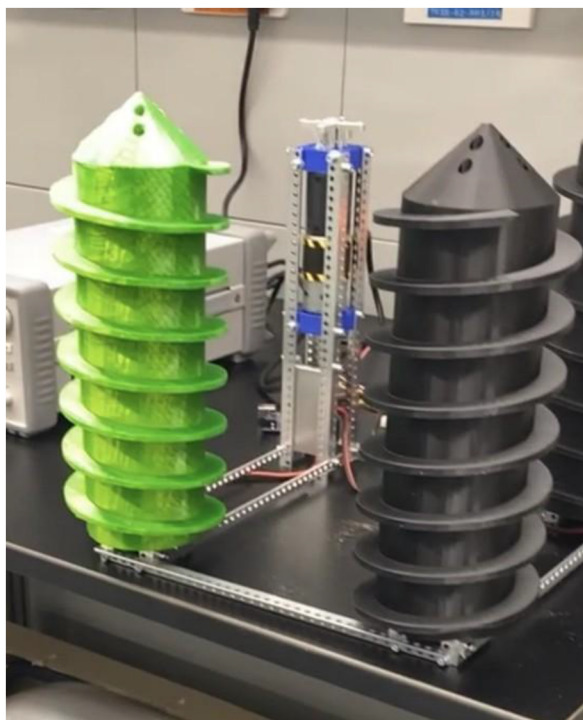


Figure 3—Leg configuration showing rotational direction, motor placement, and threading geometry

The motor shaft is directly coupled to the screw leg via a keyed internal connector to ensure efficient torque transmission. This architecture enables SUBOT to operate without external anchors, relying purely on screw thrust and symmetric torque cancellation to achieve descent and ascent.

Chassis and Frame Design

The chassis of SUBOT is built with a square frame using four aluminum L-beams (Fig.4). This configuration connects the four motor units and distributes mechanical loads evenly during burrowing. The selection of aluminum 6061 as the frame material balances weight, strength, and corrosion resistance. Finite Element Analysis (FEA) simulations show that maximum stress in the chassis during typical drilling loads remains under 40 MPa, yielding a safety factor of approximately 3 for the aluminum components. This chassis design serves not only as a mechanical backbone but also houses the embedded electronics, power delivery lines, and sensor harnesses.

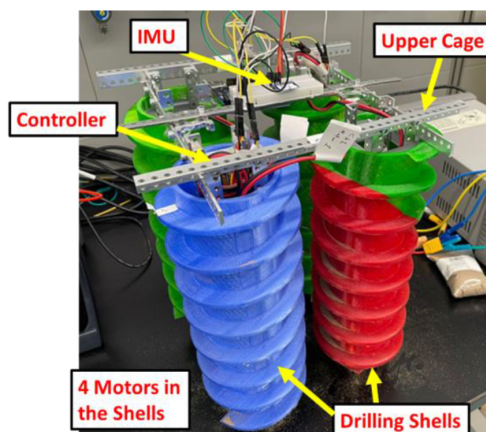


Figure 4—Full chassis assembly showing L-beams, cross-bracing, and motor integration

System Dynamics

To effectively control and evaluate the performance of SUBOT during its burrowing operation, it is essential to develop an analytical model that captures the system dynamics. The following subsections describe the mechanical forces acting on the screw-driven legs, the kinematics of vertical motion in granular media, and the combined torque and moment dynamics for the entire robotic system. These models are important for understanding motion behavior and implementing accurate control allocation and stability strategies.

Force Modeling of a Screw-Driven Leg

Each leg of the robot acts as a screw that converts rotary motion into vertical thrust. The force exerted by each leg to push the robot downward into the soil can be approximated using screw equation. The vertical thrust F_{in} generated by a single screw mechanism is related to the input torque M_{in} and the radius of the screw r as:

$$F_{in} = \frac{M_{in}}{r}$$

Where:

- F_{in} is the vertical insertion force,
- M_{in} is the torque applied by the motor,
- r is the effective radius of the screw body.

This expression assumes negligible friction loss and ideal mechanical transmission, serving as a baseline for understanding force generation in ideal conditions.

To capture the dependence of torque on depth, we introduce the concept of torsional resistance, which increases as the robot penetrates deeper due to increased contact area and compaction of sand. This is modeled by assuming a torsional spring constant K_m that scales with penetration depth z_B :

$$M_{in} = K_m z_B$$

Substituting this into the previous equation gives:

$$F_{in} = \frac{K_m z_B}{r}$$

This relation shows that the vertical force required increases linearly with depth, reflecting the growing resistance of the surrounding medium. It also emphasizes the need for high-torque actuation to sustain burrowing at deeper layers.

Kinematic Model of Vertical Displacement

The conversion of motor angular speed into vertical motion is governed by the pitch of the screw and the slippage factor encountered in loose sand. In the case of a single-start screw (i.e., one thread per rotation), the lead L is equal to the pitch.

The ideal vertical speed V_z of the robot as a function of motor rotational speed ω_i is:

$$V_z = \frac{L\omega_i}{360}$$

This formula assumes perfect mechanical contact and zero slip, which is not representative of real-world conditions in granular environments. To account for slippage, we introduce a factor $k_s \in [0,1]$, where $k_s = 0$ indicates no slippage and $k_s = 1$ indicates total slip.

The adjusted vertical speed becomes:

$$V_z = \frac{(1 - k_s)L\omega_i}{360}$$

Integrating this velocity over time gives the penetration depth z_B :

$$z_B = \frac{(1 - k_s)L\omega_i t}{360}$$

Substituting back into the force equation, we derive the insertion force in terms of motor speed and time:

$$F_{in} = \frac{K_m \left(\frac{(1 - k_s)L\omega_i t}{360} \right)}{r}$$

This relation provides a dynamic model for estimating how force and torque requirements evolve during the burrowing process, taking into account both the mechanical parameters of the screw and environmental factors like slippage.

System-Level Torque and Moment Dynamics

SUBOT consists of four screw legs, each contributing both to the total vertical force and to the torque/moment balance of the robot. To model the full actuation system, we define the control input vector:

$$u = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}$$

Where ω_i is the rotational speed of the i^{th} leg.

The combined effects of these legs on thrust force T , and roll moment M_r , and the pitch moment M_p can be expressed as:

$$\begin{bmatrix} T \\ M_r \\ M_p \end{bmatrix} = B \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix}$$

Where the control allocation matrix B is derived based on robot geometry, leg position, and moment arm lengths l . For a symmetrical cross-shaped layout, the matrix can be written as:

$$B = \begin{bmatrix} \frac{K_m(1 - k_s)Lt}{360r} & \frac{K_m(1 - k_s)Lt}{360r} & \frac{K_m(1 - k_s)Lt}{360r} & \frac{K_m(1 - k_s)Lt}{360r} \\ l \frac{K_m(1 - k_s)Lt}{360r} & l \frac{K_m(1 - k_s)Lt}{360r} & -l \frac{K_m(1 - k_s)Lt}{360r} & -l \frac{K_m(1 - k_s)Lt}{360r} \\ -l \frac{K_m(1 - k_s)Lt}{360r} & l \frac{K_m(1 - k_s)Lt}{360r} & -l \frac{K_m(1 - k_s)Lt}{360r} & -l \frac{K_m(1 - k_s)Lt}{360r} \end{bmatrix}$$

This matrix maps the rotational speeds of the legs to net vertical thrust and orientation moments. It forms the core of the control allocation strategy that could be developed in the future work and for future work also the coupling effect and nonlinear terms could be considered in the model.

This model provides the basis for distributing control efforts across actuators in a way that both propels the robot downward and corrects for unwanted tilt. It also enables compensation for uneven terrain or local variations in soil resistance, which can lead to torque imbalance if not accounted for.

Control Architecture

Maintaining a stable orientation during vertical burrowing is essential for accurate deployment of subsurface sensors and for ensuring that the robot does not deviate from its intended trajectory. Given the soft and dynamic nature of granular media, any imbalance in motor torque or small asymmetry in soil properties can result in unwanted pitch or roll, which may lead to failure in drilling or inability to ascend.

To address this, SUBOT is equipped with a closed-loop feedback control system based on dual-axis Proportional–Integral–Derivative (PID) controllers. These controllers work in parallel to stabilize the robot's

pitch and roll during both descent and ascent. The control loop utilizes real-time orientation data from an Inertial Measurement Unit (IMU), specifically the GY-521 (MPU6050), which combines a 3-axis accelerometer and a 3-axis gyroscope to calculate the robot's roll and pitch angles.

The control logic is implemented on an ATmega328P microcontroller (Arduino Uno), which reads orientation data at fixed intervals and updates motor commands accordingly.

Control Loop Overview

The system architecture consists of the following key components shown also in Fig. 5:

- **Sensor Module:** GY-521 IMU continuously senses the robot's orientation (pitch and roll angles).
- **Microcontroller:** ATmega328P processes the sensor data and computes control errors. Also provide PID controller for pitch (X-axis), and another for roll (Y-axis).
- **Motor Drivers:** Two 2x40A motor controllers receive PWM signals and regulate motor power accordingly.

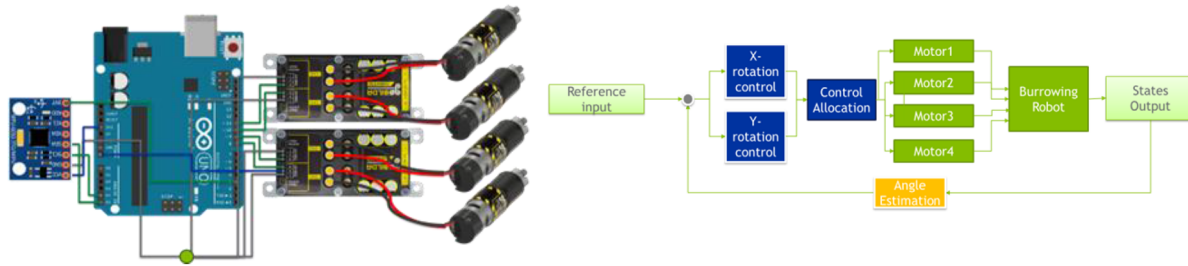


Figure 5—System wiring (Left) and Block diagram (Right) showing IMU → PID Controller → Control Allocation → Motor Outputs

PID Controller Design

The primary goal of each PID controller is to minimize the tilt angle error with respect to a level reference (i.e., 0° pitch and 0° roll). Each PID loop operates on the following control law:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

Where:

- $u(t)$ is the control effort (output command to motor allocation),
- $e(t)$ is the error between desired and measured orientation angle,
- K_p, K_i, K_d are the proportional, integral, and derivative gains, respectively.

Initial testing used gains:

- $K_p = 5$
- $K_i = 0.2$
- $K_d = 0.005$

However, experimental observations showed that this configuration led to excessive overshoot and instability, particularly in granular conditions where tilt disturbances accumulate quickly. As a result, the integral gain K_i was set to zero, leading to significantly improved performance and responsiveness. The final adopted controller parameters were:

- $K_p = 5$
- $K_i = 0$
- $K_d = 0.005$

This turned the control system into a PD controller, removing the accumulated error term that was found to cause sluggish or unstable behavior during transient phases of motion.

Software Implementation and Timing

The entire control loop runs on every one second. The Arduino Uno reads the IMU via I²C, computes the orientation angles using complementary filtering, and feeds these into the PID loops.

After computing the PID outputs, the control allocation routine computes the individual motor speeds and sends PWM signals to the motor drivers. In this way, the robot can continuously adjust its posture while descending into or ascending from the sand.

The simplicity of the hardware-software interface enables low-cost, replicable deployment and minimizes latency, which is critical for systems operating in high-uncertainty media like loose sand.

Experimental Evaluation

The functionality of the SUBOT system was validated through a series of controlled laboratory experiments aimed at evaluating its mechanical stability, control effectiveness, and burrowing capability. This section presents the setup and results of these tests, focusing particularly on how different PID gain configurations affect the robot's ability to self-stabilize while penetrating a loose sand medium.

The objectives of the experiments were:

- To assess the robot's ability to burrow vertically in granular media.
- To evaluate the effectiveness of the PID-based orientation control in correcting tilt disturbances.
- To determine the impact of integral gain on system stability and recovery performance.

Experimental Setup

A sandbox environment measuring approximately 1.1 m × 0.5 m × 0.5 m was constructed and filled with uniformly graded dry sand to replicate typical unconsolidated soil conditions. The sand was leveled manually and compacted lightly.

SUBOT was powered via a bench-top power supply delivering 15V at a maximum current of 7A. The system was programmed with the Arduino-based control logic described in Section 4, and data acquisition was performed using serial communication to log orientation data in real time (pitch and roll angles) during each test.

Each experimental run began with SUBOT placed at the surface of the sandbox in an upright position. The robot was then commanded to initiate the burrowing process. Motor rotational directions and PID controller gains were defined in the microcontroller code prior to each run. The depth of penetration was monitored visually, and orientation correction performance was evaluated using time-series plots of pitch and roll using the IMU data.

Trail A– With Integral Gain

In the first trial, the PID controller was configured with the following gains:

- Proportional Gain $K_p = 5$
- Integral Gain $K_i = 0.2$

- Derivative Gain $K_d = 0.005$

As Fig. 6 and Fig. 7 shows during descent, the robot encountered small pitch disturbances of approximately 20° , introduced naturally by slight inconsistencies in sand compaction. The controller was expected to recognize this tilt and correct it within a few seconds.

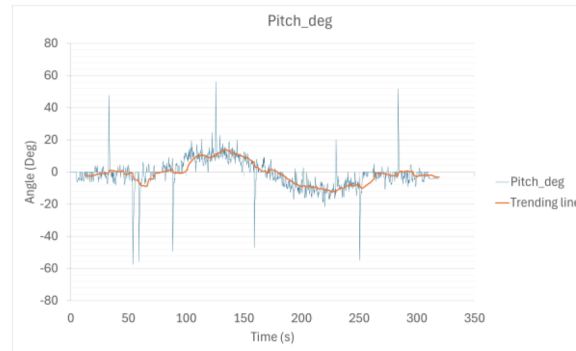


Figure 6—Pitch angle vs. time

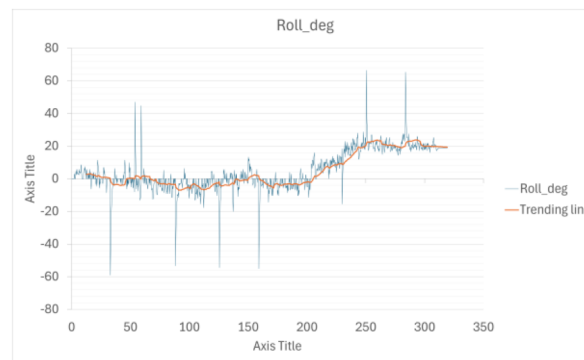


Figure 7—Roll angle vs. time showing oscillatory and unstable behavior

Observations:

- The robot was unable to recover from the 20° pitch angle.
- Rather than stabilizing, it oscillated around the disturbed orientation.
- Correction efforts were sluggish and delayed, likely due to the accumulation of integral error.
- Roll control showed similar instability, with growing deviation instead of convergence.

These results suggest that the integral term caused the control system to lag behind fast disturbances, leading to degraded performance.

Trail B – With Integral Gain Disabled

In the second trial, the controller gains were adjusted:

- Proportional Gain $K_p = 5$
- Integral Gain $K_i = 0$
- Derivative Gain $K_d = 0.005$

This configuration effectively converted the controller into a PD system, eliminating the accumulation of error over time, the result is shown in Fig. 8 and Fig. 9.

Observations:

- The robot successfully recovered from the 20° initial tilt within ~3 seconds.
- Pitch angle smoothly converged toward 0°, with minimal overshoot.
- Roll angle correction was similarly stable, and the robot maintained a near-vertical trajectory during descent.
- The robot was able to reach a depth of approximately 75 cm without deviating significantly from vertical alignment.

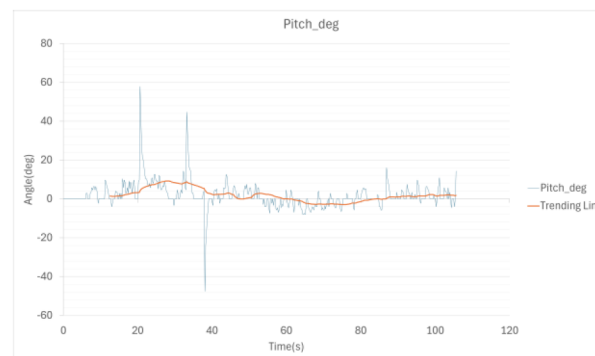


Figure 8—Pitch angle vs. time with $K_i = 0$, showing successful recovery

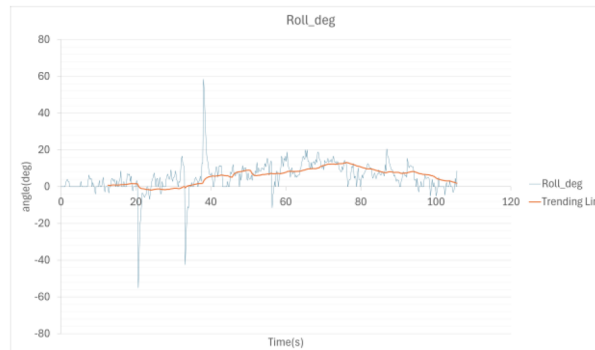


Figure 9—Roll angle vs. time with $K_i = 0$, demonstrating improved convergence

The second test showed clear superiority in terms of control response, highlighting that integral action was unnecessary and possibly detrimental for a system operating in highly dynamic environments like sand.

Summary of Experimental Findings and Discussion

The findings in Table. 1 demonstrate the critical importance of PID tuning in burrowing systems. In particular, environments with high variability and low mechanical damping — such as loose sand — benefit more from proportional-derivative control than from full PID implementations with non-zero integral gain. The results validate the design assumptions behind SUBOT's control system and confirm that, under appropriate tuning, the robot can successfully burrow, self-level, and return to the surface autonomously.

Table 1—Summary of Trail A and Trail B parameters and results.

Trial	PID Gains (P, I, D)	Recovery from Tilt	Max Stable Depth	Notes
A	5, 0.2, 0.005	No	< 30 cm	Unstable correction, overshoot
B	5, 0, 0.005	Yes	~75 cm	Smooth recovery, minimal overshoot

The experimental results clearly show that SUBOT is capable of burrowing to depths of up to 75 cm in dry sand while maintaining a near-vertical orientation, without the need for external anchoring or guidance systems. A key factor in this performance is the robot's quad-leg screw propulsion system, which provides balanced thrust and torque-canceling capabilities. By rotating the legs in a diagonally opposing pattern (two CW, two CCW), the system maintains a net-zero yaw moment, similar in concept to quadcopter drones. This configuration allows the robot to drill downward without rotating its own body, preserving orientation and ensuring efficient propulsion.

In terms of control, the switch from a full PID controller to a PD controller (by setting the integral gain K_i to zero) significantly improved stability and responsiveness. Integral action, while beneficial in systems requiring long-term error elimination, was found to be counterproductive in the context of sand drilling, where transient disturbances dominate and control authority must respond quickly. This insight aligns with observations in other robotic domains where derivative action is favored in high-variability or low-damping systems.

These findings not only validate the design choices made in SUBOT's control strategy but also offer a foundation for tuning similar systems in future development efforts.

Conclusion

This work presents the design, modeling, control, and experimental validation of SUBOT, a robotic system developed for deploying exploration seismic sensors beneath the surface of loose sand. Addressing key limitations in current seismic sensor deployment methods, SUBOT combines mechanical robustness, high-torque propulsion, and real-time feedback control into a scalable platform suitable for geophysical and petroleum applications.

Mechanically, the robot features a quad-leg screw-driven propulsion system that provides thrust while canceling net torque through a counter-rotating diagonal configuration. The structural integrity of the chassis and motor mounts was validated through finite element analysis, ensuring a safety factor of approximately 3 under operating conditions. This structural design enables SUBOT to reach depths of up to 75 cm in loose sand.

From a control standpoint, the integration of dual-axis PID controllers, operating in real time on an ATmega328P microcontroller, enables precise orientation correction during descent and ascent. The control system was refined through experimental analysis, revealing that disabling the integral gain significantly improved the robot's responsiveness and overall stability. The final control configuration successfully maintained pitch and roll near 0° during descent, which is critical for sensor placement fidelity and robot recovery.

Experimental results demonstrated that SUBOT can operate in unstructured sandy environments without the need for external tethers, anchors, or manual intervention. This represents a significant leap forward in the development of subsurface autonomous systems. The robot's design is particularly well-suited for seismic sensor deployment, where accurate placement and orientation are critical to ensuring data quality and reducing noise. These features position SUBOT as a promising candidate for future deployment in seismic monitoring, environmental sensing, and possibly planetary exploration where similar subsurface access is required.

Future work will focus on extending the robot's capabilities to operate in more complex or heterogeneous soil types, integrating sensor payloads, and enhancing autonomy through adaptive slip estimation, and multi-robot coordination. The long-term vision includes deploying SUBOT-type robots in large numbers to create dynamic, reconfigurable sensor networks in challenging environments with minimal human effort.

In conclusion, this work demonstrates that subsurface robotic systems can be both practical and scalable, offering a foundation for transformative applications in geosciences, energy exploration, and beyond.

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